Milestone: Logical Model: Relational Model
Group 4

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IE 6700 Data Management for Analytics

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Percentage of Effort

Contributed by Student1: 50%

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Logical Model: Relational Model

CUSTOMER

ATTRIBUTE	TYPE	CONSTRAINTS
Customer_id	Int	Primary key, not null
Name	Varchar	Not null
Gender	Char(1)	Null
Date_of_birth	Date	Null
Phone_number	Varchar	Unique, not null
Address	Varchar	Null
Account_type	Varchar	Not null

RESTAURANT

ATTRIBUTE	TYPE	CONSTRAINTS
Restaurant_id	Int	Primary key, not null
Name	Varchar	Not null
Cuisine_type	Varchar	Not null
location	Varchar	Not null
rating	Decimal(2,1)	Null

MENU

ATTRIBUTE	TYPE	CONSTRAINTS
Menu_id	int	Primary key, not null
Restaurant_id	Int	FK- restaurant_id, not null
Effective_date	Date	Not null

MENUITEM

ATTRIBUTE	TYPE	CONSTRAINTS
Item_id	Int	Primary key, not null
Menu_id	Int	FK – menu_id, not null
Name	Varchar	Not null
Price	Decimal(6,2)	Not null
Availability	Boolean	Default true

ORDER

ATTRIBUTE	TYPE	CONSTRAINTS
Order_id	Int	Primary key, not null
Customer_id	Int	FK – customer_id, notnull
Restaurant_id	Int	FK -resturant_id, not null
Promo_id	Int	FK – promo_id, null allowed
Order_time	Datetime	Not null
Total_amount	Decimal(8,2)	Not null
Status	Varchar	Not null

ORDER_ITEM

ATTRIBUTE	TYPE	CONSTRAINTS
Order_item_id	Int	Primary key, not null
order_id	Int	FK- order_id, not null
Item_id	Int	FK – item_id, not null
Quantity	Int	Not nul
Item_price	Decimal(6,2)	Not null

PROMOTION

ATTRIBUTE	TYPE	CONSTRAINTS
Promo_id	Int	Primary key, notnull
Code	Varchar	Unique, not null
Discount_type	Varchar	Not null
Value	Decimal(5,2)	Not null
Validity	Date	Not null

PAYMENT

ATTRIBUTE	TYPE	CONSTRAINTS
Payment_id	Int	Primary key, not null
Order_id	Int	FK – order_id, not null
Method	Varchar	Not null
Transaction_id	Varchar	Unique, not null
Amount	Decimal(8,2)	Not null
Status	Varchar	Not null

DRIVER

ATTRIBUTE	TYPE	CONSTRAINTS
Driver_id	Int	Primary key, not null
Name	Varchar	Not null
Rating	Decimal(2,1)	Null
Contact	Varchar	Unique, not null
Status	Varchar	Not null

DELIVERY

ATTRIBUTE	TYPE	CONSTRAINTS
Delivery_id	Int	Primary key, not null
Order_id	Int	FK – order_id, not null
Driver_id	Int	FK – driver_id, not null
Pickup_time	Datetime	Not null
Droff_time	Datetime	Null
Eta	Datetime	Null
Status	Varchar	Not null

Description of Revised EER Model

1. **Customer:** Stores customer details. One customer can place many orders (**1:N** with Order).
2. **Restaurant:** Holds restaurant info. A restaurant can have many menus and receive many orders (**1:N** with Menu, Order).
3. **Menu:** Represents a restaurant's list of dishes for a specific time. Each menu belongs to one restaurant and has many menu items (**1:N**).
4. **Menu_Item:** Lists individual dishes under a menu. Each item belongs to one menu but can appear in many orders through **Order_Item** (**M:N** resolved by Order_Item).
5. **Order:** Central entity linking customer, restaurant, promotion, payment, and delivery. Each order has many items, one payment, and one delivery; promotion is optional.
6. **Order_Item:** Connects orders and menu items. Stores quantity and item price (**M:N** bridge between Order and Menu_Item).
7. **Promotion:** Stores discount details. One promotion can apply to many orders, but each order uses at most one (**1:N**).
8. **Payment:** Contains payment details for each order. One-to-one with Order (**1:1**).
9. **Driver:** Holds driver information. A driver can complete many deliveries (**1:N** with Delivery).
10. **Delivery:** Tracks order delivery details (pickup, drop-off, ETA). Linked **1:1 with Order** and **N:1 with Driver**.